1. Project Title: StudyGenie: AI-Powered Smart Study Planner

2. Project Statement and Outcomes:

- The AI-powered Smart Study Planner is an intelligent tutoring system designed to assist students in constructing a personalized daily study plan and generating quizzes for effective learning. The project leverages AI to analyze the student's syllabus, subject preferences, current grade, and time availability to create structured daily study tasks. Additionally, the system will generate daily quizzes to reinforce learning and track progress. A dynamically generated daily study plan based on their syllabus and available time. Auto-generated quizzes tailored to the topics covered each day. A structured, goal-oriented approach to learning that adapts to student progress.
- The AI tutor will ensure that students stay on track, effectively manage their study time, and enhance retention through interactive quizzes. The project will conclude with a fully functional web application, comprehensive testing, and detailed documentation for future enhancements.

3. Modules to be Implemented:

- User Authentication & Profile Setup Module
- AI-based Study Plan Generation Module
- Quiz Generation Module
- Dashboard Development Module
- Notifications Module
- Testing, Review, and Documentation

4. Week-wise Module Implementation and High-Level Requirements:

Weeks 1-2: User Authentication & Profile Setup Module

- Implement user authentication (registration and login).
- Allow students to enter their study details(Title,Syllabus, Date range,Subject,Current Grade)
- Develop a module to extract structured data from uploaded syllabus files.
- Preprocess and organize the syllabus into manageable topics.

Weeks 3-4: AI-Based Study Plan and Quiz Generation Module

- Implement AI algorithms to generate a **daily study plan** based on syllabus topics, time availability, and difficulty level.
- Develop an automated quiz generation module that creates daily quizzes based on completed topics.
- Ensure quiz difficulty adapts to the user's performance.

- Implement a progress tracking system that updates dynamically based on user input.

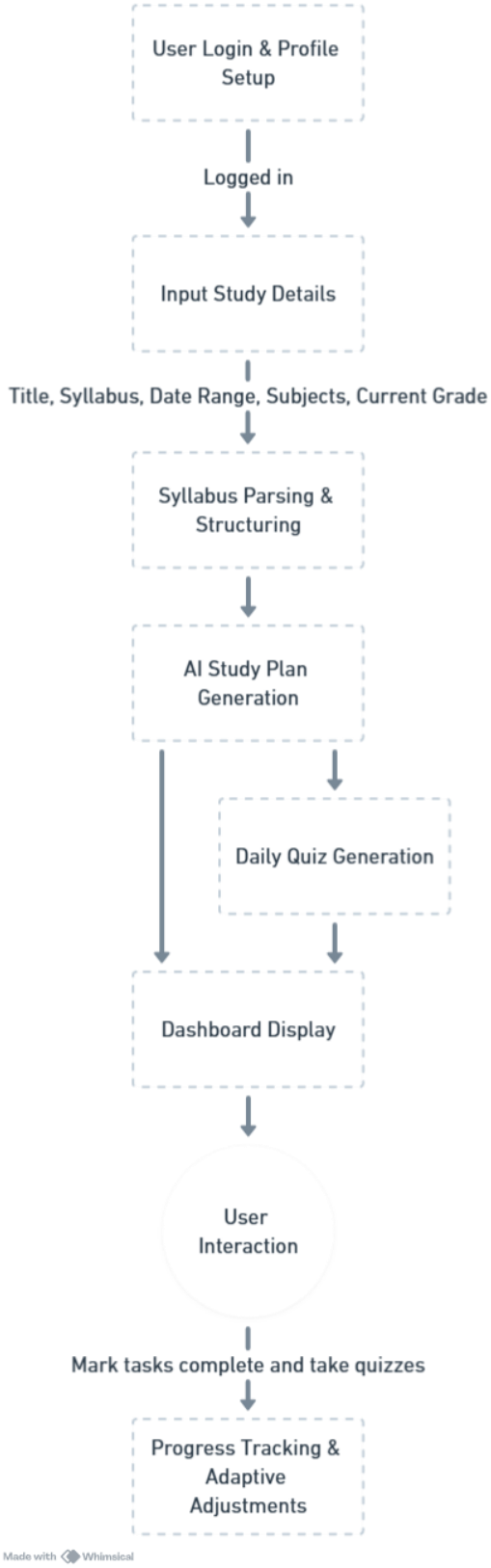
Weeks 5-6: Dashboard Development & Notifications Module

- Design and develop an intuitive dashboard to display(Daily study tasks, Upcoming quizzes, and Progress tracking graphs)
- Ensure users can mark study tasks as completed and take quizzes directly on the platform.
- Send Email reminders & notifications to the user on his progress & day's topic

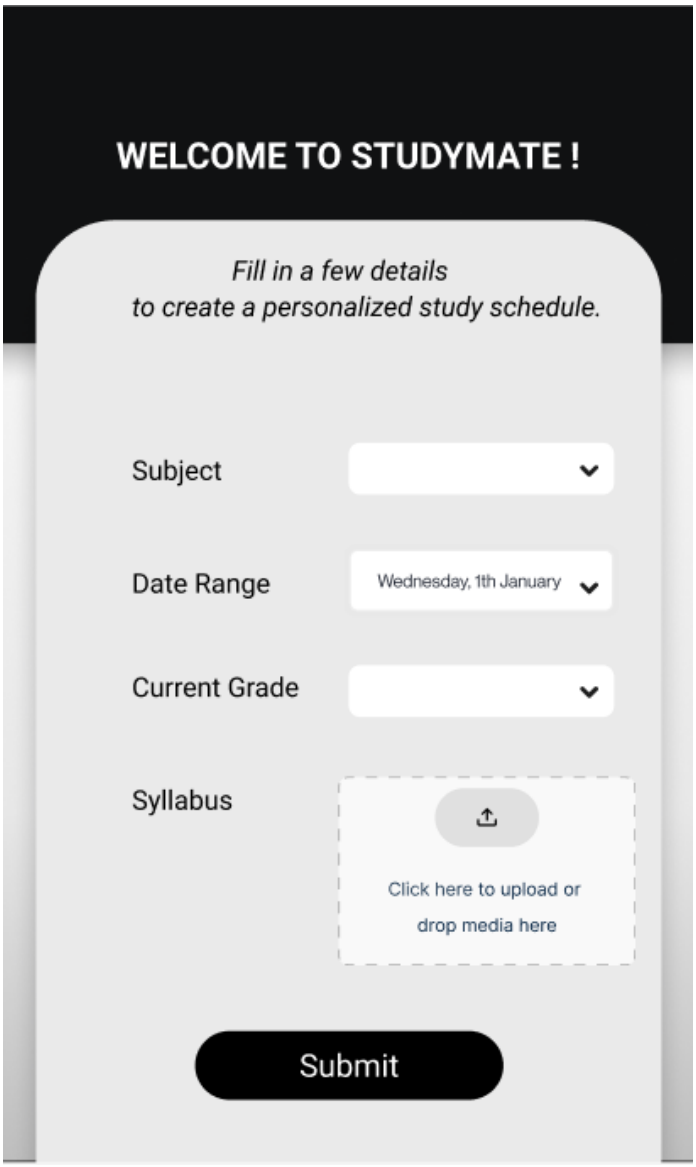
Weeks 7-8: Testing, Review, and Documentation

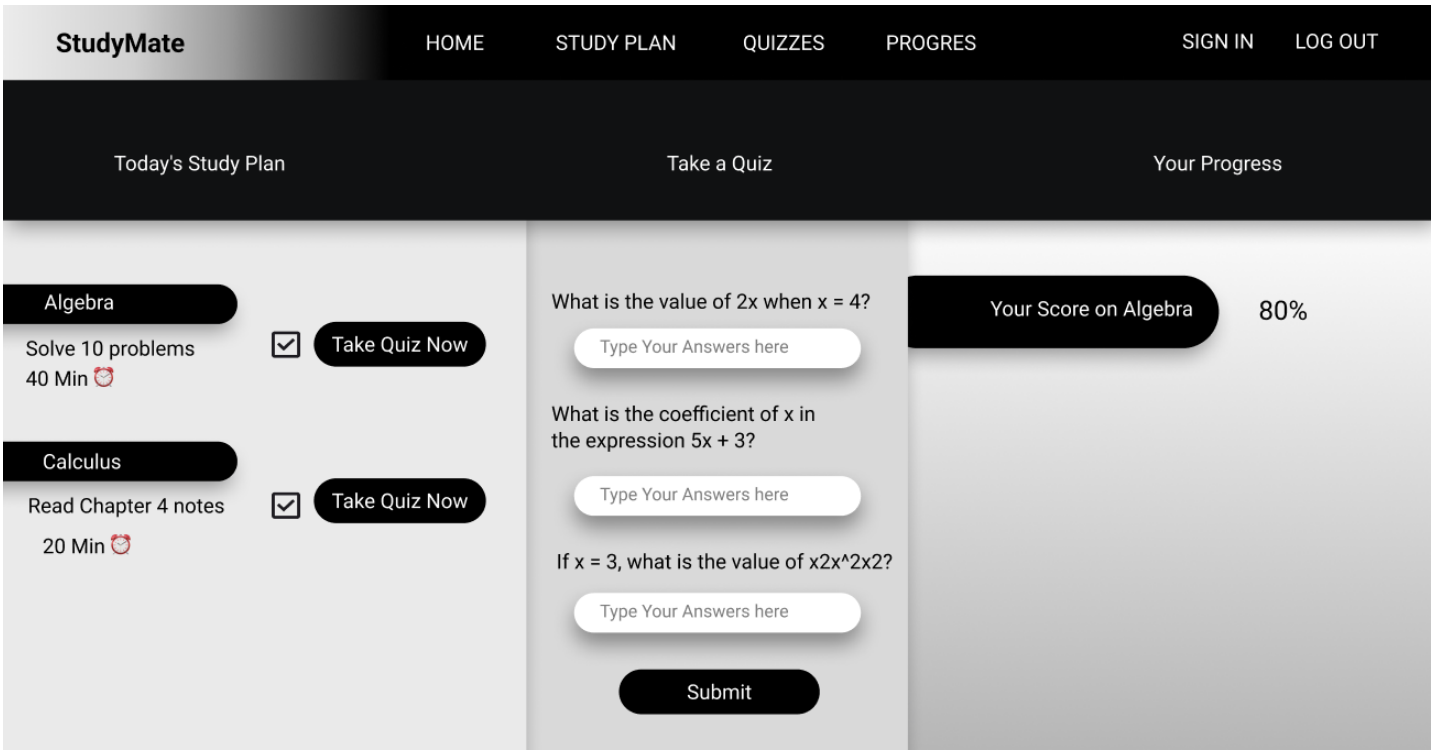
- Conduct a thorough review of the entire system, including functionality, security, and user interface.
- Address any identified bugs or issues and perform necessary fixes.
- Prepare comprehensive documentation & presentation covering system architecture, user guides, and technical specification

5. Flow chart:


User Login & Profile Setup
Logged in
Input Study Details
Title, Syllabus, Date Range, Subjects, Current Grade
Syllabus Parsing & Structuring
AI Study Plan Generation
Daily Quiz Generation
Dashboard Display
User Interaction
Mark tasks complete and take quizzes
Progress Tracking & Adaptive Adjustments
Made with Whimsical

6. Sample output:




StudyMate
HOME
STUDY PLAN
QUIZZES
PROGRES
SIGN IN
LOG OUT
Today's Study Plan
Take a Quiz
Your Progress
Algebra
Solve 10 problems
40 Min
Take Quiz Now
Calculus
Read Chapter 4 notes
20 Min
Take Quiz Now
What is the value of 2x when x = 4?
Type Your Answers here
What is the coefficient of x in the expression 5x + 3?
Type Your Answers here
If x = 3, what is the value of x2x^2x2?
Type Your Answers here
Submit
Your Score on Algebra
80%